Customer Behavior Analysis

Retail purchase data → revenue drivers, segments, targeted actions

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Project Snapshot

Scope

3900 purchases, 18 columns

Goal

Translate data into insights & actions

Tools

Python • MySQL • Power BI

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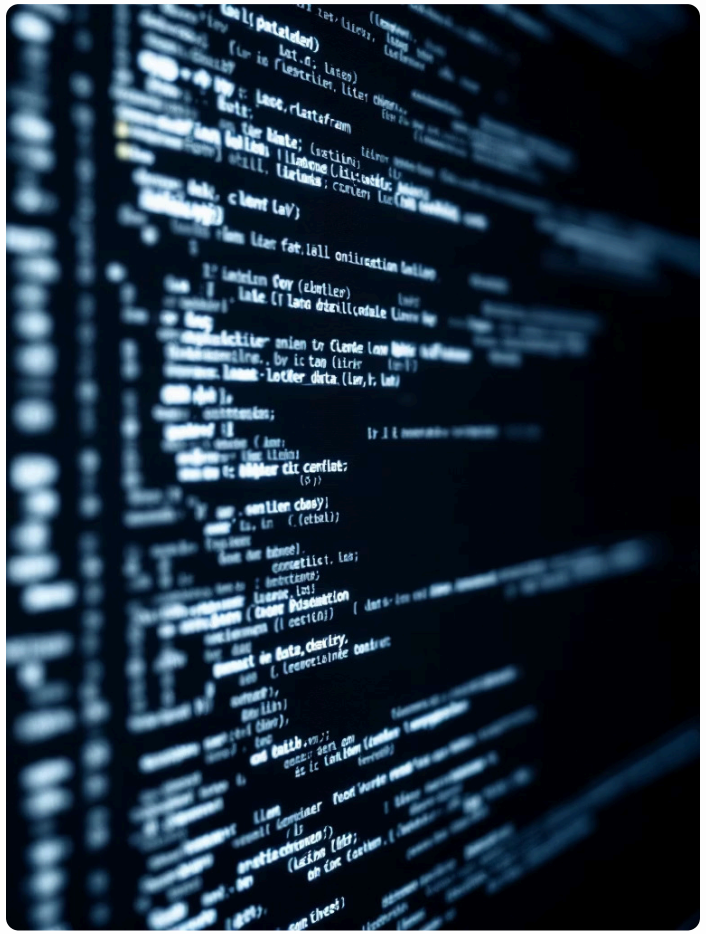
Data & Preparation



Key steps

- Imputed 37 missing review ratings (median by category)
- Renamed columns to snake_case
- Engineered age_group & purchase_frequency_days
- Dropped redundant promo_code_used

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Exploratory Stats

Summary Statistics

	Silver	Sleac	Median	Mean	Median	Minimum	Minimum	Maximum	Maximum
Mean	15.8%	10.8%	10.3%	18.4%	10.5%	10.8%	20.9%	150.0%	150.9%
Lout mear	16.5%	·	5.3%	4..0%	45.9%	10.0%	5.7%	5.9%	2.7%
Laange fination	12.5%	·	10.1%	40.5%	30.8%	10.7%	10.5%	5.7%	3.5%
Minimage mertal	11.2%	·	22.7%	86.4%	66.9%	8.9%	8.7%	6.5%	1.6%
Protical State	12.5%	·	20.9%	35.4%	20.9%	10.9%	11.7%	1.7%	7.6%
Pterinal Campart	32.7%	·	22.7%	25.4%	26.9%	20.0%	15.9%	5.5%	0.5%
Maximum Parrnall	12.5%	·	10.0%	200%	10.3%	20.0%	17.7%	17.5%	6.2%
Fat Cate	13.7%	·	21.7%	30.3%	36.7%	40.0%	20.7%	2.5%	3.5%
Aucrmemaerl Sommid	17.5%	·	16.5%	0..7%	56.5%	20.6%	17.7%	0.5%	0.6%
Tabalis	201.0%	30.7%	265.7%	20.04%	200.0%	410.0%	27.7%	35.5%	7.3%

Customers

Count 3900 • Age mean 44

Spending

Avg purchase $59.76 • Std $23.69

Ratings

Mean 3.75 (37 missing imputed)

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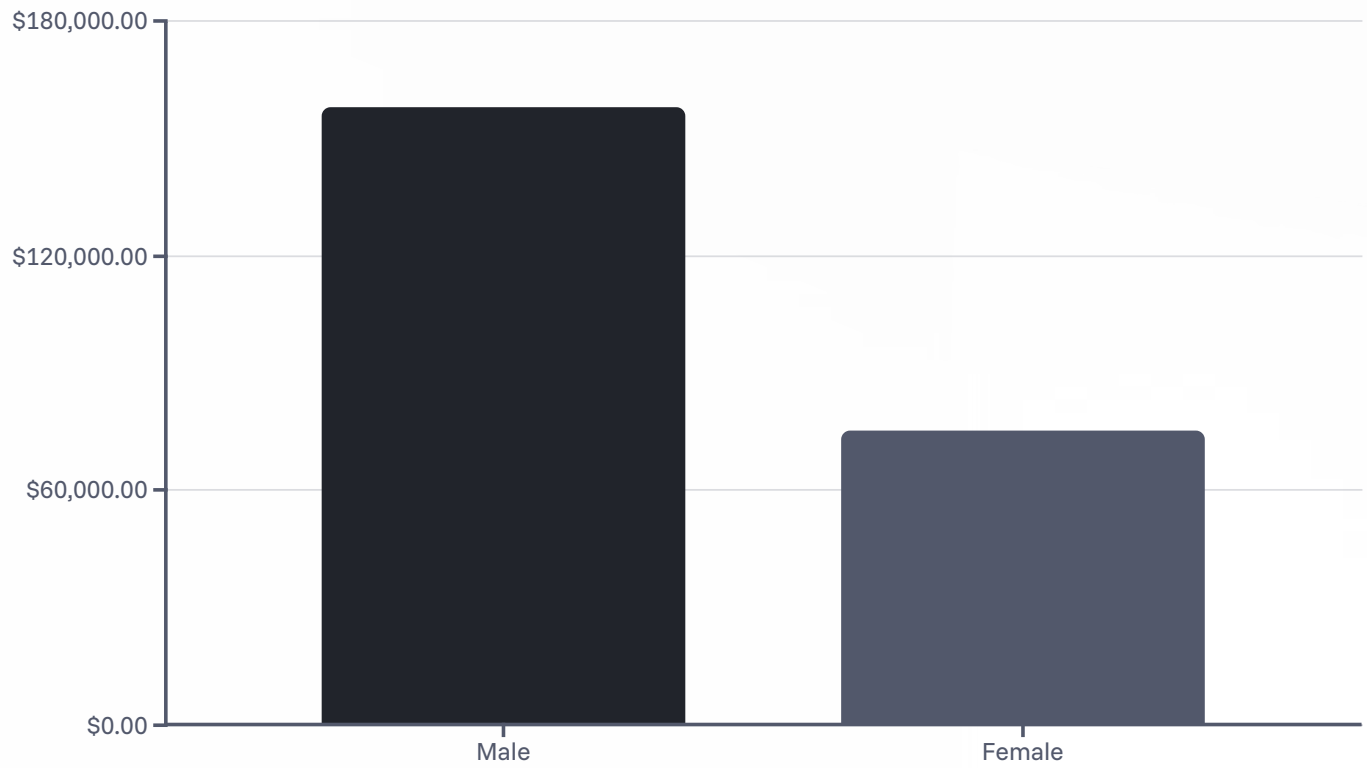
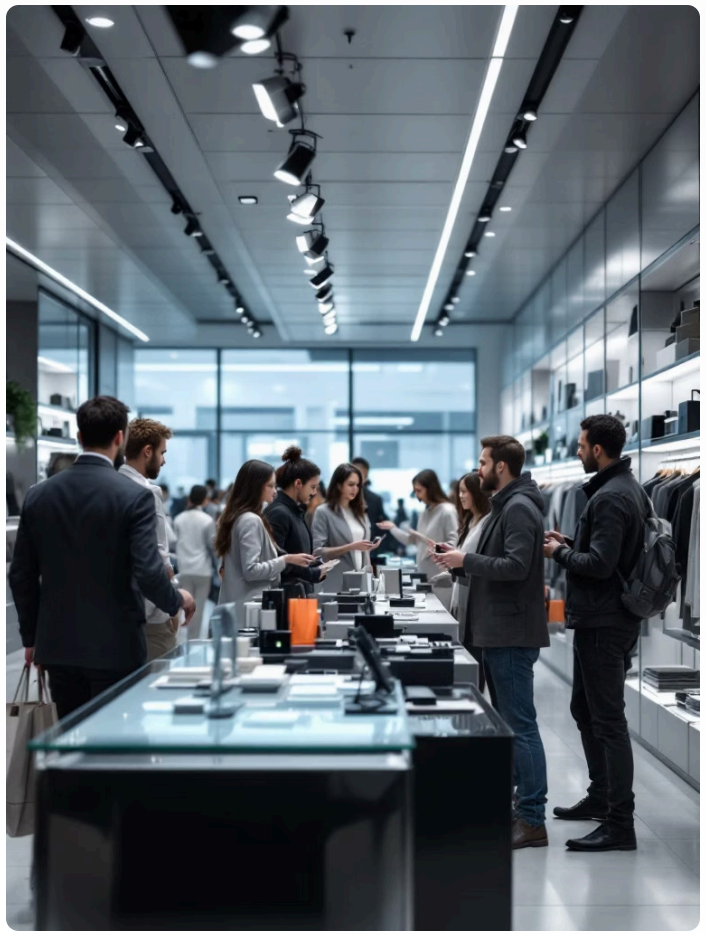
Revenue by Segment



Male customers generate higher total revenue despite similar avg spend



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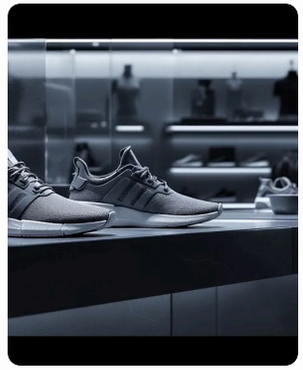
Product & Discount Insights

Hat

50% discount purchases

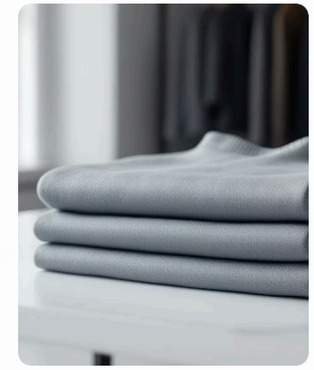


Sneakers

49.66% discount purchases

Coat

49.07% discount purchases



Sweater

48.17% discount purchases

Pants

47.37% discount purchases

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Top Products & Ratings

Gloves

Avg rating 3.86

Sandals

Avg rating 3.84

Boots

Avg rating 3.82

Other top-rated: Hat 3.80 • Handbag 3.78

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Customer Segments & Loyalty

3.9K
Customers

$59.76
Avg Purchase

3.75
Avg Rating

Segments: New 83 • Returning 701 • Loyal 3116

Subscribers: Yes revenue $62,645 • No revenue $170,436

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Power BI Dashboard Highlights

Key Metrics

3.9K customers • $59.76 avg spend • 43% discounts

Category Revenue

Clothing 100K • Accessories 70K • Footwear 30K • Outerwear 10K

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Recommendations

Focus

Prioritize Clothing & Accessories (top revenue)

Retention

Strengthen subscriptions to convert repeat buyers

Discounts

Targeted discounts to protect margins

Experience

Improve delivery & service to lift ratings

Segmentation

Personalize offers by age and behavior

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